1 May 2026
Bright Learn
SQL Aggregate Functions & Operators
Ex 3

① SELECT DISTINCT department
FROM students;

students

department
IT
HR
Finance

② ~~SELECT FROM students~~
SELECT AVG (age) AS avg_age, department
FROM students
GROUP BY department;

students

department	age
IT	20 5
HR	22
Finance	23

③ SELECT COUNT (*), department AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;

students

department	student_id
IT	2
HR	2

④ SELECT age
FROM students
WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

⑤ SELECT department, student_id, name, age
FROM students
WHERE department = ('IT' OR 'HR')
AND age > 21;

student_ID	name	age	department
2	Bob	22	HR
4	Diana	23	Finance
5	Eve	22	HR

⑥ SELECT SUM (credits), department AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5.

~~SUM~~ department	total_credits
IT	11

⑦. SELECT course_id, course_name, department, credits
FROM courses
WHERE credits != 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

⑧ SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

course_id	course_name	credits
102	Python	~~3~~ 4
103	Data Science	4
~~105~~	~~Statistics~~	~~3~~
101	SQL Basics	3

(9) SELECT MAX (grade) AS max_grade,
MIN (grade) AS min_grade,
AVE (grade) AS ave_grade
FROM enrollments;

max_grade	min_grade	ave_grade
98	78	84.6

(10). SELECT COUNT (course_id) AS enrollment_count
FROM enrollments
GROUP BY enrollment_count;

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

(11). SELECT SUM (salary) AS total_salary,
SUM (bonus) AS total_bonus, department
FROM salaries
GROUP BY department;

department	total_salary	total_bonus
IT	112 000	10 500
HR	109 000	7500
Finance	70 000	6000

(12) SELECT AVE (salary) AS ave_salary,
departments
FROM salaries
~~WHERE ave_salary~~ GROUP BY department
HAVING AVE (salary) > 55 000;

department	ave_salary
IT	61 000
Finance	~~54 500~~ 70 000

(13) ~~SELECT employee_id, name, salary, bonus, total_compensation~~
~~FROM salaries~~
~~SELECT sum (salary)~~ SELECT sum (salary + bonus) AS
total_compensation,
employee_id, name, salary,
bonus
HAVING sum (salary + bonus) > 60 000;

employee_name	name	salary	bonus	total_compensation
1	Jon	60000	5000	65000
3	Spike	70000	6000	76000
4	Jyke	62000	5500	67500

(14). SELECT SUM(budget) AS total_budget,
AVE(budget) AS avg_budget,
department, FROM projects
GROUP BY department HAVING AVE(budget) > 70 000;

department	total_budget	ave_budget
IT	270 000	135 000
Finance	80 000	80 000

(15) SELECT project_id, project_name, department, budget
FROM projects
~~GROUP BY budgets BETWEEN 50 000 AND 130000~~
~~WHERE budget~~ WHERE budget BETWEEN 50 000 AND 130 000
AND department != 'Marketing'

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80 000
~~3~~	~~Dashboard~~	~~IT~~	~~150~~
4	Website	Marketing	60 000
5	HR Portal	HR	50 000